T20 WORLD CUP – BEST PLAYING XI USING DATA ANALYTICS & POWER BI

Final Project Report

Prepared By: Sanchit Akhare
Tool Used: Power BI
Domain: Sports Analytics
Project Type: Data Visualization & Decision Analytics

Executive Summary

This project aimed to build the most balanced and competitive T20 World Cup Playing XI using a performance-driven selection framework instead of subjective player choices.

A structured analytical model was created in Power BI to evaluate players across multiple dimensions including batting efficiency, strike rate, bowling effectiveness, economy, consistency, finishing capability, and match impact.

Role-based filters and qualification rules were applied to identify players who collectively satisfy two major objectives:

- Build a team capable of scoring 180+ runs consistently
- Build a bowling attack capable of defending 150 runs

The final outcome was a balanced squad with aggressive opening capability, stable middle order, explosive finishing power, multiple bowling options, and efficient specialist bowlers.

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1. Project Objective

The purpose of this project was to:

- Build the Best T20 Playing XI using statistical analysis.
- Eliminate selection bias.
- Compare players using measurable KPIs.
- Create a scalable sports analytics framework.

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2. Dataset Overview

Dataset includes player-level T20 World Cup performance data across:

- Batting Statistics

- Bowling Statistics
- Strike Rate
- Batting Average
- Bowling Average
- Bowling Strike Rate
- Economy Rate
- Dot Ball %
- Balls Faced
- Boundaries
- Playing Role

Data was segmented into:

1. Power Hitters / Openers
2. Anchors / Middle Order
3. Finishers
4. All Rounders
5. Specialist Fast Bowlers

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3. Methodology

Players were shortlisted using role-specific thresholds.

Openers

Selection variables:

- Batting Average
- Strike Rate
- Boundary %
- Batting Position

Anchors

Selection variables:

- Batting Average
- Strike Rate
- Average Balls Faced

Finishers

Selection variables:

- Strike Rate
- Lower-order impact
- Secondary bowling ability

All Rounders

Selection variables:

- Batting Strike Rate
- Economy
- Bowling Strike Rate

Fast Bowlers

Selection variables:

- Economy
- Bowling Average
- Dot Ball %
- Bowling Strike Rate

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4. Dashboard Analysis & Interpretation

Dashboard Page 1 – Power Hitters / Openers

Key Findings

- Openers generated:
    - Strike Rate: 117.08
    - Batting Average: 20.16
    - Boundary Contribution: 50.34%
    - Average Balls Faced: 13.65

Insights

- Jos Buttler demonstrated elite consistency with average and strike balance.
- Rilee Rossouw delivered explosive acceleration.
- Alex Hales maintained strong attacking output.
- Quinton de Kock showed highest scoring efficiency.

Conclusion:
Top-order strategy prioritizes aggressive starts while maintaining wicket stability.

Dashboard Page 2 – Anchors / Middle Order

Key Findings

Top performers:

- Virat Kohli

- Suryakumar Yadav
- Glenn Phillips
- Daryl Mitchell
- Lorcan Tucker

Insights

- Virat Kohli delivered exceptional consistency.
- Suryakumar Yadav provided the highest attacking middle-order intent.
- Glenn Phillips balanced stability and acceleration.

Conclusion:
Middle order is designed to absorb pressure while maintaining scoring momentum.

Dashboard Page 3 – Finishers

Key Findings

Top performers:

- Glenn Maxwell
- Marcus Stoinis
- Sikandar Raza
- Hardik Pandya
- Curtis Campher

Insights

- Sikandar Raza delivered the strongest balance between batting and bowling.
- Maxwell added finishing acceleration.
- Hardik contributed dual impact.

Conclusion:
Lower order improves both batting depth and tactical flexibility.

Dashboard Page 4 – All Rounders

Key Findings

Combined Metrics:

- Economy: 7.32
- Bowling Strike Rate: 18.43

Top Performers:

- Shadab Khan
- Mitchell Santner

- Rashid Khan
- Sikandar Raza

Insights

- Shadab Khan emerged as the strongest wicket-taking all-round option.
- Santner offered economical control.
- Rashid added attacking spin.

Conclusion:
All-rounders significantly increased team adaptability.

Dashboard Page 5 – Specialist Fast Bowlers

Key Findings

Combined Metrics:

- Economy: 7.32
- Bowling Average: 22.50
- Bowling Strike Rate: 18.43
- Dot Ball %: 40.26%

Top Performers:

- Sam Curran
- Anrich Nortje
- Shaheen Shah Afridi
- Tim Southee

Insights

- Anrich Nortje showed elite wicket efficiency.
- Sam Curran delivered strong overall control.
- Shaheen added powerplay dominance.

Conclusion:
Bowling selection emphasized wicket-taking while restricting scoring.

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5. Final Team Performance

Overall Team Metrics:

- Strike Rate → 154.54
- Batting Average → 39.60
- Average Balls Faced → 19.71
- Economy → 6.47

- Bowling Strike Rate → 13.09
- Bowling Average → 14.12
- Dot Ball % → 41.15%

Interpretation:

The final XI exceeds expected benchmarks and achieves balance between aggressive scoring and defensive capability.

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6. Key Business & Analytical Insights

- Data-driven selection reduces subjective decision-making.
- Multi-role players improve team flexibility.
- Strike Rate alone does not guarantee selection.
- Economy and Bowling Strike Rate together better predict bowling impact.
- Balanced teams outperform star-heavy teams.

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7. Limitations

- Historical tournament data only.
- Match conditions not included.
- Opposition strength excluded.
- Venue conditions excluded.

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8. Future Enhancements

Future versions may include:

- Match simulations
- ML-based player prediction
- Venue-adjusted selection
- Win probability analysis
- Dynamic squad recommendation

Final Conclusion

This project successfully demonstrated how Power BI and sports analytics can transform cricket decision-making into a measurable and objective process.

The final selected XI combines explosive batting, stable middle-order execution, efficient finishing, adaptable all-round contributions, and high-impact bowling performance.

The dashboard proves that data analytics can be used to create stronger team selection frameworks and improve strategic decisions in competitive sports.

END OF REPORT